

StudySync

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OVERVIEW

Welcome to your new study buddy, StudySync! If you've ever found yourself buried under a mountain of books with a to-do list longer than your arm, this app is here to save the day. Picture this: a tool that not only helps you manage your study time but does so with flair! With a sprinkle of algorithms, a dash of color, and a commitment to keeping you on track, this app transforms your study chaos into a streamlined, organized plan—making studying feel less like a chore and more like an adventure!

KEY FEATURES

User Input for Subjects

Users can input the name, difficulty (on a scale of 1 to 5), and deadlines for multiple subjects. Input validation ensures that difficulty levels are within the specified range.

Subject Prioritization

Subjects are prioritized based on their difficulty using a greedy algorithm, ensuring that more challenging subjects receive appropriate attention.

Study Hours Allocation

The program allocates study hours and minutes to subjects based on their difficulty and ensures that the total allocated time does not exceed available study hours. Adjustments are made to allocate extra time to subjects with upcoming exams.

Optimization Algorithms

- **Hungarian Algorithm:** Used for optimizing the assignment of subjects to study tasks based on a cost matrix that factors in difficulty and deadlines.
- **Knapsack Algorithm:** Implements a classic dynamic programming approach to maximize the difficulty achieved with available study hours.

Study Plan Generation

The project generates a daily study plan, displaying allocated hours and minutes for each subject based on their upcoming exam days. The plan is dynamically adjusted, ensuring that subjects with imminent exams receive extra study time.

User-Friendly Menu Interface

An interactive menu allows users to easily navigate through options to input subjects, optimize their study plan, and view daily allocations.

CONCLUSION

Say goodbye to study stress and hello to your new StudySync! This project is not just a tool; it's your quirky sidekick in the journey of academic excellence. With its witty algorithms and engaging features, it turns the daunting task of studying into a fun and organized experience. So, grab your favorite snacks, fire up it, and let's tackle those subjects like the nerds we are! Your future self will thank you for it—preferably with a well-deserved victory dance!